

# Lecture 2: Microprogrammed Control Unit

## 1. Introduction to Microprogrammed Control Unit (MCU)

- A **Microprogrammed Control Unit** generates control signals using a **program stored in memory**, called **control memory**.
- Unlike Hardwired CU, it uses **microinstructions** to control the operation of the CPU.
- Microprogrammed control is **flexible** and easier to design for complex instruction sets.

### Key Idea:

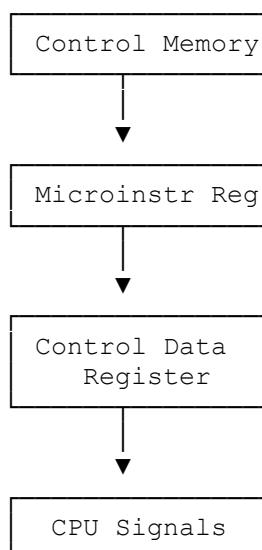
- The control signals for executing machine instructions are **stored as sequences of microinstructions**.
- Each microinstruction specifies **control signals** for one step of instruction execution.

## 2. Structure of Microprogrammed Control Unit

### Components:

1. **Control Memory (CM):**
  - Stores **microinstructions**.
  - Can be **ROM** (read-only) or **RAM** (programmable).
2. **Microinstruction Register (MIR):**
  - Holds the current microinstruction being executed.
3. **Control Address Register (CAR):**
  - Points to the **next microinstruction** in the control memory.
4. **Control Data Register (CDR):**
  - Holds the **control signals** generated by the microinstruction.
5. **Sequencer:**
  - Determines the **address of the next microinstruction**.
  - Supports branching, looping, or conditional execution.
6. **Timing and Clock Signals:**
  - Synchronize microinstruction execution.

### Diagram (Simplified):



## 3. Microinstruction Format

**Microinstructions** specify which control signals should be **activated** in a single step.

### Two Types of Microinstructions:

1. **Horizontal Microinstruction:**
  - Each **bit directly controls a hardware signal**.
  - Provides **maximum parallelism**.
  - Requires **more bits** per instruction.
2. **Vertical Microinstruction:**
  - Encodes control signals in **binary fields**.
  - Requires **fewer bits**.
  - Needs a **decoder** to generate actual signals.

### Example:

For instruction `ADD R1, R2`:

- Microinstruction sequence might be:
  1. Read R1 and R2 into ALU.
  2. Perform addition in ALU.
  3. Store result back into R1.

Each step is a microinstruction stored in control memory.

## 4. Operation of Microprogrammed Control Unit

### Steps:

1. **Fetch microinstruction from Control Memory**
  - CAR points to the next microinstruction.
2. **Load microinstruction into MIR**
  - MIR holds it temporarily.
3. **Activate Control Signals**
  - CDR generates the necessary signals to CPU hardware.
4. **Determine next microinstruction**
  - Sequencer updates CAR based on:
    - Next sequential address
    - Branch (conditional/unconditional)
5. **Repeat until machine instruction completes**

### Key Advantage:

- Changing instruction behavior requires **updating microprogram** rather than hardware redesign.

## 5. Advantages and Disadvantages

### Advantages:

1. **Flexibility:** Easy to modify or add instructions by updating control memory.
2. **Easier to design and debug:** Reduces complex logic circuit design.
3. **Supports complex instruction sets (CISC).**

### Disadvantages:

1. **Slower** than hardwired control (microinstruction fetch adds overhead).

2. Requires **more memory** for storing microinstructions.
3. Less suitable for very high-speed CPUs.

## 6. Comparison with Hardwired Control Unit

| Feature                   | Microprogrammed CU                | Hardwired CU                       |
|---------------------------|-----------------------------------|------------------------------------|
| Control Signal Generation | From microinstructions in memory  | From combinational logic circuits  |
| Speed                     | Slower                            | Faster                             |
| Flexibility               | High – easy to modify/update      | Low – hardware redesign required   |
| Complexity                | Simple for large instruction sets | Complex for large instruction sets |
| Instruction Set           | Supports complex CISC easily      | Suitable for simple RISC           |

**Mnemonic:** “Microprogrammed is memory-driven and flexible, Hardwired is logic-driven and fast.”

## 7. Microprogramming Techniques

1. **Sequential Microprogramming:**
  - Microinstructions executed sequentially.
  - Simple, no conditional branching.
2. **Conditional Microprogramming:**
  - Microinstructions can **branch** based on condition flags.
  - Useful for instructions like `JUMP`, `BRANCH` in CPU.
3. **Parallel Microprogramming (Horizontal):**
  - Multiple control signals activated simultaneously.
  - Efficient for hardware utilization.

## 8. Summary / Key Points

- Microprogrammed CU uses **control memory** to generate signals.
- Microinstructions simplify CPU design and allow **flexible instruction set updates**.
- Horizontal vs. Vertical microinstructions differ in **bit-width and parallelism**.
- Slower than Hardwired CU but **ideal for complex instruction sets**.
- Modern CPUs often use **hybrid approaches**: mostly microprogrammed for complex control, but hardwired for high-speed paths.